Matching Starbursts Part 2

The game is simple. Reach your hand into the bag and choose a Starburst (bag has 4 red and 2 yellow). Return the first Starburst into the bag, mix up the bag, and reach in for another. If the color matches the first one, you win! What is the probability of winning?

with replacement

1. Are the two events "1st is Red" and "2nd is Red" independent? Explain.

Yes. Knowing whether or not the 1st is Red does not change the probability that the 2nd is red.

2. For independent events P(A and B) = P(A) * P(B). Find the following probabilities (as fractions).

P(1st is Red and 2nd is Red) = $\frac{4}{6} \cdot \frac{4}{6} = \frac{16}{36}$

P(1st is Yellow and 2nd is Yellow) = $\frac{2}{6} \cdot \frac{2}{6} = \frac{4}{36}$

P(winning) = $\frac{16}{36} + \frac{4}{36} = \frac{20}{36} = 0.56$

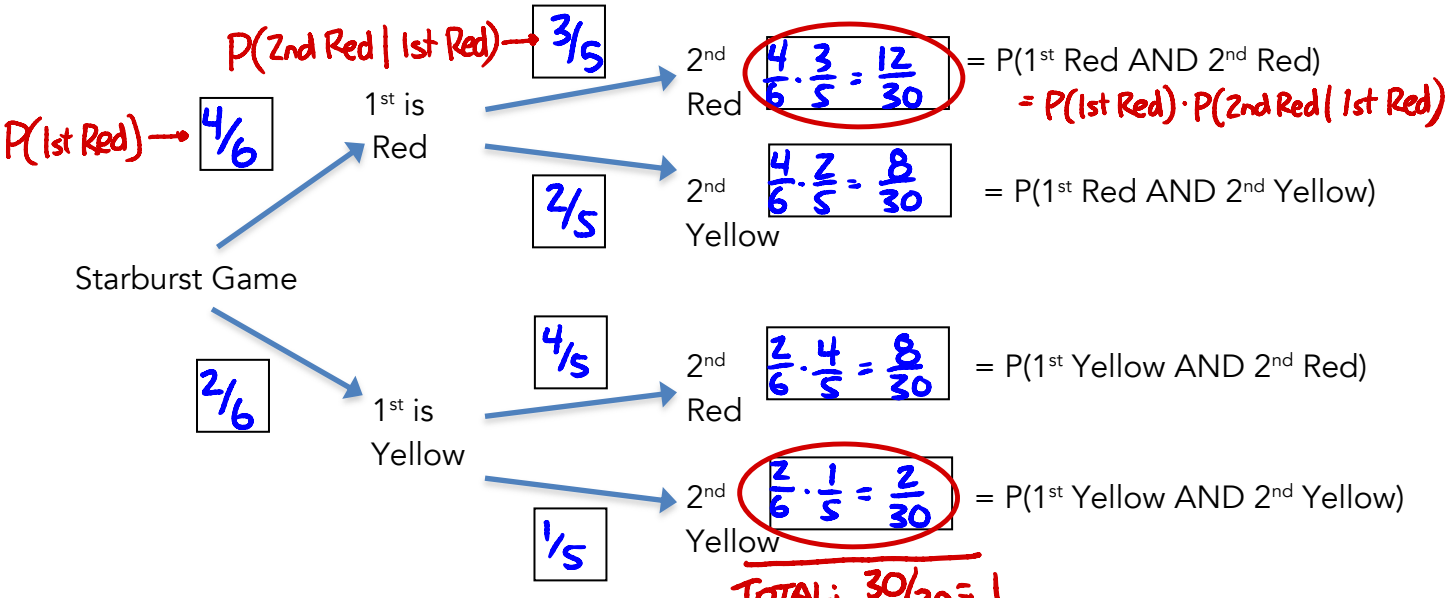
STOP

3. Now a twist! This time you will NOT replace the 1st Starburst before getting the 2nd. Your group should play the game 10 times total. Record your number of wins at the front of the room. Based on the results from your whole class, what is the estimated probability of winning this game?

$\frac{39}{80} = 0.49$

without replacement

4. Let's try to use a Tree Diagram to calculate the theoretical probability.
Fill in the blank boxes with the correct probabilities (as fractions).



5. Find the theoretical probability of winning this game.

$\frac{12}{30} + \frac{2}{30} = \frac{14}{30} = 0.47 = P(\text{win})$

6. What is the probability that the 1st candy was red, given that the person won the game?

$\frac{12}{14} = 0.86$ $P(\text{1st red} \mid \text{win}) = \frac{P(\text{1st red AND win})}{P(\text{win})} = \frac{12/30}{14/30} = 0.86$

Lesson 3.6 – Tree Diagrams

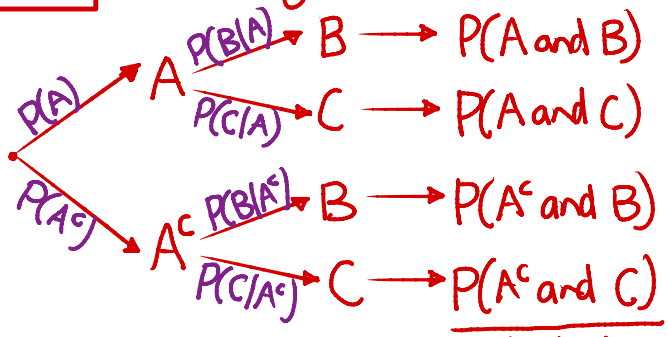
QuickNotes

LT#1 General Multiplication Rule

$P(A \text{ and } B) = P(A) \times P(B|A)$ ("given that"; $P(B|A)$ is a conditional probability)

* If A, B are independent *
→ $P(B|A) = P(B)$, so
→ $P(A \text{ and } B) = P(A) \cdot P(B)$

LT#2 Tree Diagrams

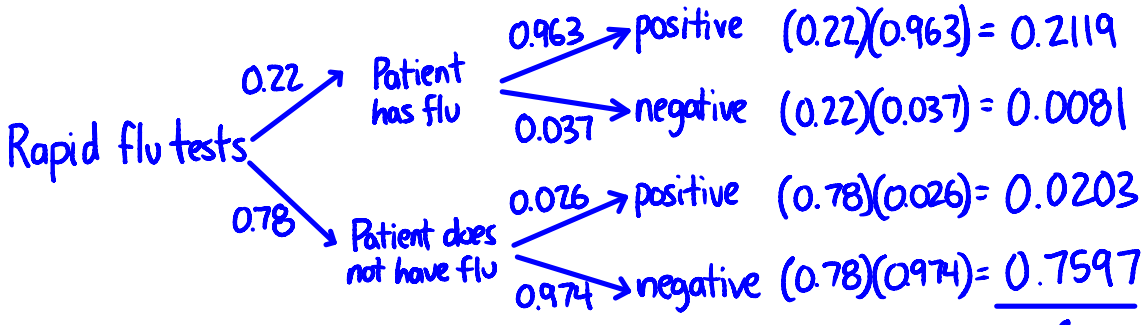


* P(at least 1) = 1 − P(none)

Check Your Understanding

Many urgent care facilities offer rapid flu tests that will give results in 15 minutes or less. Suppose that for a certain week in November, the urgent care center in Kentwood, Michigan is visited by many people who get rapid flu tests. For these patients, suppose that 22% of them do in fact have the flu. The rapid flu test will incorrectly give a negative result for 3.7% of the patients that do have the flu and incorrectly give a positive result for 2.6% of the patients that don't have the flu.

1. Construct a tree diagram to model this chance process.



2. Find the probability that a randomly selected patient gets a positive result.

$0.2119 + 0.0203 = 0.2322$

3. If we select 4 patients at random, what is the probability that at least one of them gets a positive result?

$P(\text{at least 1}) = 1 - P(\text{none})$
$= 1 - (0.7678)^4 = 0.6525$

4. Given that a randomly selected patient is a positive result, how likely is it that they actually have the flu?

$\frac{0.2119}{0.2322} = 0.913$ $\quad P(\text{have flu} \mid +) = \frac{P(\text{have flu AND } +)}{P(+)}$